EVALUATING VISUAL AND TEXTUAL FEATURES FOR PREDICTING USER 'LIKES'

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ABSTRACT

Computationally modeling users 'liking' for image(s) requires understanding how to effectively represent the image so that different factors influencing user 'likes' are considered. In this work, an evaluation of the state-of-the-art visual features in multimedia understanding at the task of predicting user 'likes' is presented, based on a collection of images crawled from Flickr. Secondly, a probabilistic approach for modeling 'likes' based only on tags is proposed. The approach of using both visual and text-based features is shown to improve the state-of-the-art performance by 12%. Analysis of the results indicate that more human-interpretable and semantic representations are important for the task of predicting very subtle response of 'likes'.

Index Terms— Likes, Feature Representation, Tags, Recommendation, Image

1. INTRODUCTION

Why does a user 'like' a (set of) image(s)? Answering this question requires understanding factors associated with images that influence the user to express preference for some images over the other [1]. Given only one image that a user 'likes', it is very difficult to answer this question. But if a collection of images that the user 'liked' are given, modeling user's taste and inference on "What 'kind' of images the user 'likes'?" can be made. Using this model, images that the user has not seen and might 'like' can be recommended. And conversely, given a set of images and users, we should be able to predict the user(s) who 'liked' the image.

While 'like' can be defined as "finding something enjoyable and satisfactory", it is a very subtle phenomenon which is very difficult to be formalized by methods and processes. There has been a lot of research in the areas of emotion recognition, aesthetic assessment and sentiment prediction [2, 3, 4]. And the response of 'likes', even though more subtle, is influenced by all these factors [5] and more (objects present, their perspectives/poses, scenes, colorfulness etc.). The rich repository of tags, comments and titles (in Social Media) also contribute to inferring people's opinions, emotions and other semantic responses to model user 'likes'.

Presence of such multiple multimodal representations necessitates an evaluation of how these different visual and text-based features (carrying different levels of semantic information) perform at the task of predicting user 'likes'. We do this by comparing their performance at predicting the right user given a set of images and vice versa. Based on these experiments we analyse why certain features perform better and try to answer why.

Given images (in Fig 1) that come in the first page of images liked (or tagged as 'favorite') by a random user on Flickr, by looking at the images, we can say that the user likes images with lots of people in broad day light. Now, given just the tag cloud associated with the same images, we can infer that the user likes street photography, mostly shot in New York, Manhattan and London. These tags also convey that the user likes concepts such as people, women, real love etc.

So, we attempt to model user 'likes' using both the visual and text-based features. This is inherently different from keyword/query based image retrieval in the sense that we are trying to infer what images the users would 'like' and not just what images contain the 'concepts' defined in the tags (for example objects, scenes etc.). Even if the tags associated with images liked by different users are the same, the 'kind' of images that they 'liked' might actually be very different.

Recent studies in Psychology have shown that preference formation (or 'likes') is caused due to the interplay between how new a stimulus seems to the user and how well the user can extract the sense of it to relate it to previous knowledge [5]. Translating this into modeling users 'likes' on images:

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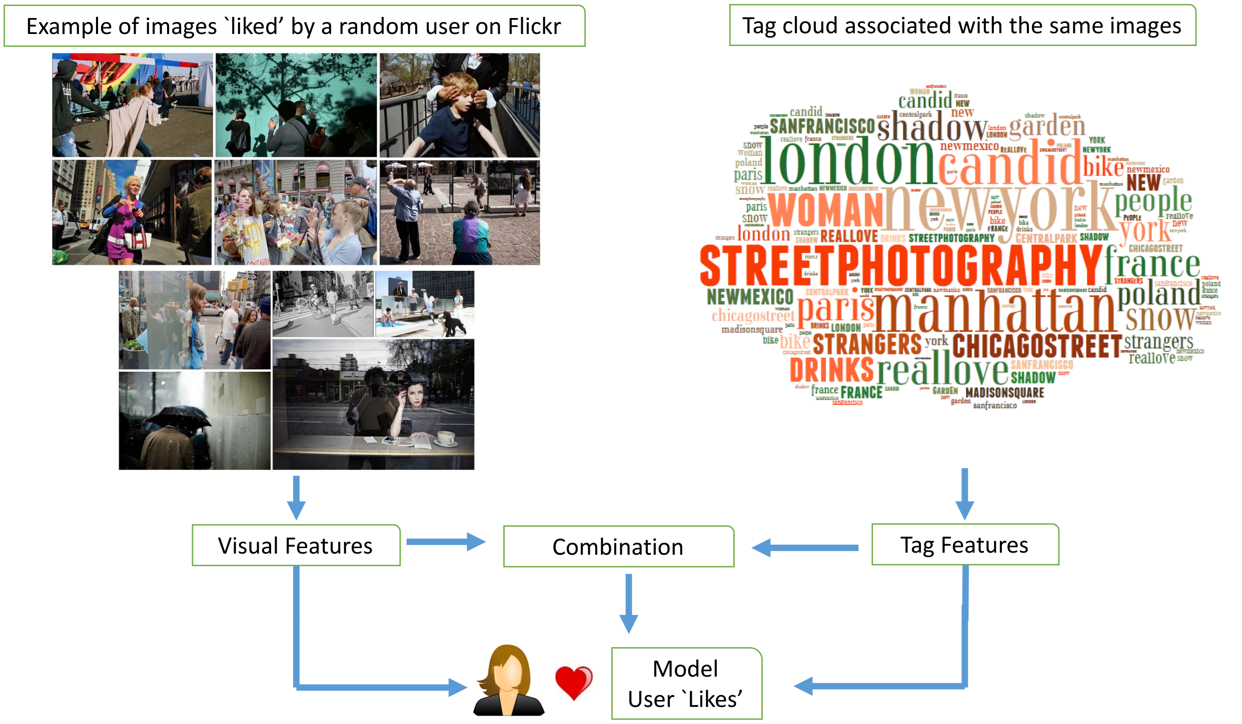

Fig. 1. Factors influencing a user's 'liking' for images can be understood both from the visual contents as well as the text associated with the images in the form of tags, titles and comments.

given a stimulus (image in this case), a user first tries to recognise the 'concepts' in the image then infers the emotional and contextual information from past experiences. This inference is then translated into a user's liking/disliking the image. There have also been studies which propose that very abstract concepts like personality of users' can also be modelled based on their 'likes' [6].

Thus, the question, which we pose at the beginning of this paper, can be more concretely answered by: a) Identifying the visual and textual associations ('concepts') that can be extracted from images, and b) Using these associations to computationally model user 'likes', given a set of images they 'liked'.

Contributions: 1) We evaluate how different image feature representations (which carry different levels of semantic information) can model user's 'likes'. Based on this evaluation, we try to give some insights into why certain features outperform others. This helps in identifying what kind of feature representations should be designed for modeling very 'subtle' subjective human responses such as user 'likes'. 2) We propose a probabilistic approach to predict user 'likes' based on tags alone (associated with images), which when combined with visual features outperforms the state-of-the-art performance by 12%.

2. DATASET AND FEATURES FOR PREDICTING USER 'LIKES'

In this section, we briefly explain the dataset, various features employed and then describe the proposed approach to predict user 'likes' based on tags.

2.1. Dataset

We constructed a dataset of images crawled from Flickr. Images tagged as favorite by 300 users (a total of 60,000 images) were crawled using the Flickr API. The associated tags, titles and comments were also crawled. The crawled images had 476,422 tags and each image had a title. From these we selected 94,497 unique tags after removing stop words like camera names, lens characteristics, and only single letter or alphabet and 'generic' words like 'photo', 'image'.

2.2. Low-Level Visual features

We use the features used in state-of-the-art [7] namely average intensity of each of the RGB color channels, number of edges, texture index, number and area of segments obtained by mean shift segmentation, objects detected by DPM, number and size of faces and GIST.

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2.3. High-Level Visual features

Motivated by the recent advancement in learning interpretable mid-level representations in images, we explore semantically meaningful features in this section, namely objects, places, style, adjective-noun pairs (SentiBank), category-level attributes and aesthetic features. For the first three features, we use the activations of Convolutional Neural Networks (pre-trained for the mentioned tasks). We used Caffe [8] to extract the features from the ReLu layer following fc7 layer, resulting in a feature of 4096 dimensions.

Objects from the Imagenet Network [9] trained on 1.3 million images from the ImageNet challenge 2012. Our goal in using these features here is to understand how object-centric features contribute to the task of predicting user 'likes'.

Places from the Places CNN [10] trained on 2.5 million images on 205 scene categories. The features were extracted in a similar manner as that of Object features described above. The goal was to see how these scene-centric features contribute to users 'likes'. We also use HybridCNN model which is pre-trained on both Places and Imagenet datasets.

Style prediction in images is a recently researched problem [11], where 80K Flickr photographs annotated with 20 curated style labels were used to finetune a pre-trained Imagenet CNN. These features are supposed to distinguish different aspects of visual style, namely photographic techniques ('HDR', 'Macro'), composition ('Minimal','Geometric'), moods ('Serene','Melancholy') etc.

SentiBank is an attribute representation based on emotion related concepts (Adjective-Noun Pairs (ANP)) [2]. We used classifiers to detect 2089 ANPs in our dataset. Adjectives are strongly related to one of the 8 emotion categories, as put forward by Plutchik [12], and nouns correspond to objects and scenes in the image. These features are helpful to model user 'likes' as often users tend to form an emotional connection with the images before 'liking' them.

Category-level Attributes [13] are useful for characterizing abstract ANPs by automatically designing attributes encoded by a compact category-attribute matrix. This is a 2000 dimensional feature vector.

Aesthetics [3] are psycho-visual statistics extracted from multiple levels, namely cell level, frame level and shot level. Though the original method was proposed on videos, it is a frame-based feature representations and thus we discard the shot level features and retain the remaining two on the images.

2.4. Tag-based Probabilistic Approach to Model User 'Likes'

While a significant role is played by the visual content in capturing user 'likes', tags associated with images often enhance the information that visual features convey and also compliment them at times. Tags primarily convey the concepts present in an image which visual detectors can also be trained to capture.

However, they often go beyond what visual features can detect - for example users may 'like' an image because of it being a place they know of or because of a person they know of, which triggers a memory from their past and enables them to have an emotional connection with the image.

In one sense, most of the tags already contain high-level semantic information like 'objects','contexts','places','affect' etc., and are so-to-say a better mid-level representation, provided we are able to find a proper way to encode them into features and use the right approach to model user 'likes'.

It should be noted that we only have information of what the user 'likes' and not what the user 'dislikes'. Therefore this can be formulated as a positive-only/implicit feedback problem where we have information about the tags of images that are 'liked' by the user.

Formally we describe the probabilistic approach for modeling user 'likes' using different text-based features by Equation 1.

$$P(like|image) = \sum_{tags} \Bigg[P(like|tag) \times P(tag|image) \Bigg] \quad (1)$$

where

$$P(like|tag) = \sum_{images} \Big\{ P(like, image|tag) \Big\}$$

where $P(tag|image)$ is the weight associated with images-tag pair (i.e., the probability of a tag appearing in an image) and $P(like, image|tag)$ is computed by taking all images having a tag and checking how many have been liked (the probability is the average over all such images).

We use the TF-IDF approach and Bayesian Personalized Ranking (BPRMF) from Implicit Feedback implemented in MyMediaLite package [14] to learn the weights on tags associated with all images and compare the performance of both. Additionally we convert the sparse tag matrix into a compressed representation using SVD (in line with [15]) and evaluate its performance.

3. EXPERIMENTS AND ANALYSIS

For the experiments, we used 200 images each, liked by 205 users (i.e., a total of 41000 images). For each user we use 100 images to train our models and the remaining as test images (i.e., a 50%/50% split on train/test data). For every user, we extract the features mentioned in Section 2 using the default parameters of the softwares provided by the cited works.

3.1. Training and Testing Protocol for Visual Features

Training

Once the features of all the images are extracted and the

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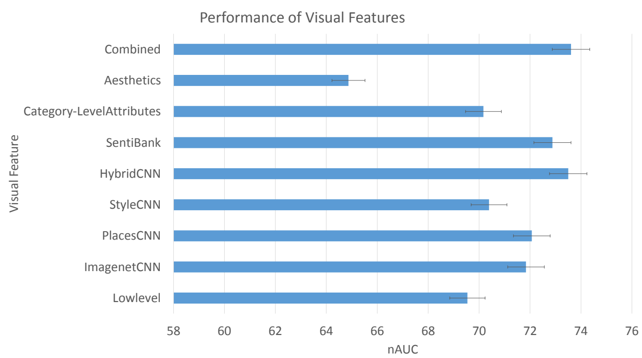

Fig. 2. Performance of Visual Features (in terms of nAUC). SentiBank and HybridCNN (pre-trained on Places+Imagenet) outperform others.

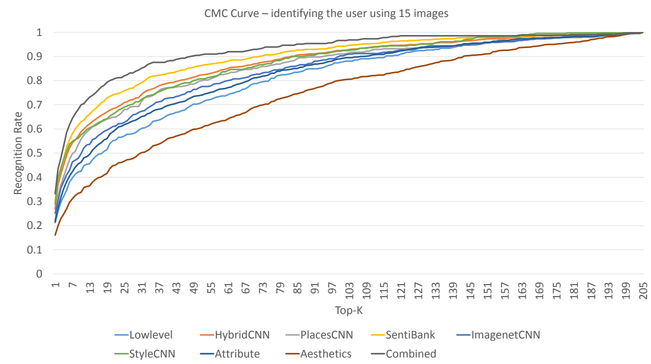

Fig. 3. Performance of Visual Features at predicting the right user, given a set of images.

dataset is divided into training and test sets, we perform a sparse regression analysis using Lasso [16]. Every training example is represented using one of the features as a vector x_n, and is associated with a output variable y_n, which is the label of the user who 'liked' it. The output is then expressed as a linear combination of input feature vectors:

$$y_n = w^T x_n \tag{2}$$

The problem can be formulated as a least square error minimization

$$E(w) = \sum_{n=1}^{N} (y_n - w^T x_n)^2 \tag{3}$$

where N $\in \{training\_set\_images\}$ and $\sum_j (|w_j| \leq \lambda)$. The regularization parameter λ is learnt over a 5-fold cross-validation setting. In order to discriminate one user from the rest of the users, we learnt one regressor for every user in a one-vs-all setting, considering training images 'liked' by that user to have a label of +1 and the remaining (i.e., images 'liked' by other users) to have a label of -1. Thus we have M regressors, where M is the number of users in our dataset. The setting is similar to that used in the state-of-the-art [7].

Testing

We use two evaluation protocols to test the performance of different features.

- Accuracy in terms of discriminating one user from the others (mean of the ROC curve for each user), which is reported as mean Area Under Curve (nAUC). This protocol helps to identify which feature is able to capture the differences between users effectively.

- Performance in identifying the user who 'liked' an image, given an image. This is different from the previous setting, in the sense that we are trying to predict which user would have 'liked' the image, ranked on the Cumulative Matching Curve (CMC), based on the traits of the image captured by each feature .

To put it in simpler terms, the first task is to predict the right images, given a user and the second task is to predict the right user, given an image.

Results and Discussion

In the first task, for all images in the test set, we obtain the regression scores β_n^m=$w^{(m)T} x_n$, where m=1, 2, ..., M users. And using the regression scores, we calculate a ROC curve for each user (which is then averaged). The results for the visual features are shown in Fig 2.

HybridCNN outperforms (73.5% nAUC with a standard deviation of 7.8%) other features. The superior performance of HybridCNN features can be explained by the fact that most of the images 'liked' by the users are either different places which can be described by concepts like 'sunny coast', 'wintering forest path', 'ski resort', 'cathedral outdoor' and so on or object centric images which can be described by 'phone booth', 'game room', 'stadium football' etc., which have been used as labels to train the HybridCNN network (on the Places and Imagenet datasets).

Intermediate network layer (ReLu followed by fc7), which has been used in our experiments, seems to be able to learn even the affective information, which is an integral part of the visual content in these images. Similar findings have been reported in [17] where network pre-trained on Imagenet gave reasonable performance on the task of visual sentiment prediction.

The next best performing feature is SentiBank detectors. The performance of this feature is intuitive, based on the findings provided in Section 1. Other features like the Low-level feature and the category-level attributes do not perform as well due to their inability to capture all the facets of the visual content which influence the user to 'like' an image.

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We wanted to investigate if users tend to 'like' positive sentiments and 'dislike' images which carry negative sentiments. For every image in the test set, a 2089 feature ANP descriptor is generated, which represents the probablities of different ANPs being present in the image. SentiBank ANPs also have an associated sentiment score in the range of [-2,2]. We took a weighted average of the sentiment score and the SentiBank descriptor to get a final score on every image, thereby getting a 'net' sentiment score on the image. However the nAUC using this approach was around 67%. Further examination of images reveled that there were users who explicitly 'liked' images carrying a negative sentiment (melancholy scenes and so on).

In the second task, given a test image x_n, we obtain the regression score β_n^m for each user m and rank all the users based on this score. We build a CMC curve, which tells us the rate at which the right user is matched in top-k ranks. The results for the visual features are shown in Fig 3.

It is interesting to note that while HybridCNN outperformed SentiBank in the first task by a small margin, in this task SentiBank outperforms HybridCNN features. A possible reason that can be ascribed to this is that, given only one image, SentiBank features are able to better capture the subjective peculiar traits in the image when compared to HybridCNN features as SentiBank feature were explicitly designed as Adjective-Noun Pairs, which directly influence a user's 'liking'/'disliking' an image. The presence of the adjectives in these ANPs might be a reason for SentiBank features to perform slightly better at this task.

One work that we did not evaluate in this section is [18], where authors propose a multi-resolution latent space, formed by Counting Grids [19] of multiple resolutions, thereby mapping similar images nearby which are then used to build one-vs-all classifiers. The input to the Counting Grid in their work are Low-Level features with some Aesthetic features. Therefore, we believe that if the multi-resolution Counting Grids are used on the features evaluated in this work, we can observe a corresponding difference in performance due the proposed approach. There was also another work which proposed deep feature representation over high level featuers [20]. However, in this work, our aim was to evaluate the efficacy of different features and not the approach in which those features are used to train the models.

3.2. Training and Testing protocol for Tag features

As described in Section 2.1, we identified 94,497 unique tags in the dataset and built a binary sparse image-tag matrix. We used this matrix to extract three feature representations namely TF-IDF, Matrix Factorization (MF) [14] and SVD representation [15].

For TF-IDF and MF, the entire matrix was used to get the weights on the image-tag pairs and on test images, the probabilistic approach in Section 2.4 was used to predict the score in line with the protocol used on testing visual features. For SVD representation, a Lasso regressor was trained on the training data and the predictions were made on the test data (similar to that of visual features). The results, in terms of nAUC are shown in Fig 4.

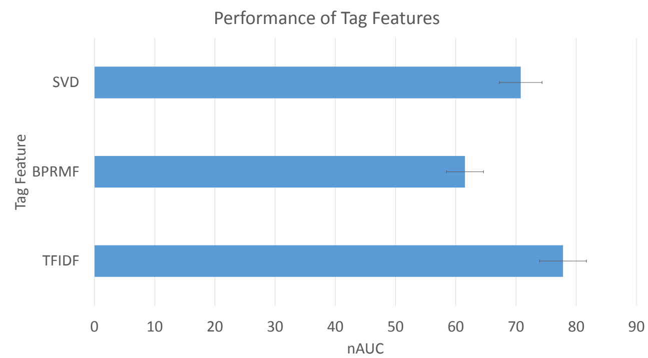

Fig. 4. Performance of Different Text Features (in terms of nAUC). TF-IDF representation outperforms others.

TF-IDF approach outperforms the other two approaches possibly due to its ability to capture the high-level semantics associated with images. MF on the other hand was designed for item recommendation and when applied to tags (as items in this context), it might not be able to perform well as this task needs the model to capture the semantics (i.e., item 'attributes') as well. SVD representation on the other hand gives comparable performance with Low-Level visual features.

It is interesting to note that the probabilistic approach on TF-IDF representation is able to outperform even the best Visual feature performance (by about 3%). This might be due to the fact that we had selected a huge vocabulary and all the tags in the test set were also ensured to belong to this vocabulary and new tags were discarded.

In the case that a new tag (which does not belong to the vocabulary) is present in the test set, distributed representations like word2vec [21] can be used to measure the semantic distance between the unknown test tag and the known tags in the vocabulary to map them accordingly.

A multimodal approach (i.e., using both text and visual features) in the late-fusion technique further boosts the performance by 4% (80% nAUC), giving about 12% improvement over the state-of-the-art (68%).

This shows that both visual and text-based features play a significant and sometimes complementary roles in capturing users' preferences enhancing the semantics of the image represented by each modality. And especially for this task of predicting user 'likes', which are very subtle, such more human-interpretable and semantic representations are important for building accurate models.

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4. CONCLUSION

In this paper, we attempted to evaluate different visual and text-based features at the task of predicting user 'likes'. Results from different experiments show that high-level features (especially SentiBank and HybridCNN, pre-trained on Places and Imagenet) outperform others, conveying the need for more human-interpretable representations for this task. A probabilistic approach was proposed to model user 'likes' only based on tags. It was seen to outperform even the visual features (by 3%) and their combination outperforms the state-of-the-art by 12%. While the results are encouraging, further investigation into combining different modalities in more effective manner, for this task of modeling user 'likes' and dealing with large set of users (say millions, in the case of real-world applications) are intriguing areas deserving further investigation.

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